Correlation Network Studio

Interactive Graph Analytics, Statistical Modeling & High-Dimensional Feature Exploration

Full-Stack System: FastAPI • SQLite • D3.js Force Physics Simulation • SciPy & NumPy Analytics

Academic Project Defense • Comprehensive Evaluation: Real Datasets, Empirical Regressions & Centrality Topology

Preface & Objective

The Challenge: High-dimensional tabular datasets obscure cross-variable correlations. Traditional N × N correlation tables scale quadratically ($O(N^2)$), overwhelming human cognitive bandwidth. Raw numerical tables hide latent clusters, hub variables, and inverse trade-offs.

Our Objective: To engineer and deploy a production full-stack visual analytics platform transforming raw numerical datasets into interactive, force-directed network graphs with linked bivariate OLS regression scatter plots and univariate distribution metrics.

The Scope: This presentation outlines our decoupled three-tier architecture (FastAPI, SQLite, D3.js), the mathematical correlation engines (Pearson & Spearman with two-tailed p-values), empirical dataset evaluations, and multi-level analytical user workflows.

SYSTEM WORKFLOW

Methodology: The Core Pipeline

Three integrated analytical stages transforming raw tabular data into interactive visual insights.

1. Data Ingestion & Storage

STEP 1: PERSISTENCE

- Aggregates and persists multi-feature datasets in SQLite relational database.
- Dynamic schema validation requires ≥ 2 numeric columns and ≥ 3 observations.
- Automated drag-and-drop CSV uploader with Pandas numeric type sanitization.
- Pre-seeded with real benchmark datasets: Academic Performance & Macroeconomics.
- UUID generation and transactional integrity guarantee reliable data retrieval.

2. Statistical Engine Core

STEP 2: ANALYTICS

- Dual correlation engine: computes full N × N Pearson and Spearman rank matrices.
- Rigorous hypothesis testing: calculates exact two-tailed Student's t-test p-values.
- Dynamic threshold filtering: strips spurious, statistically non-significant edges.
- Bivariate OLS linear regression: fits trendline, calculates slope, intercept, and R^2.
- Univariate statistics: NumPy frequency histogram binning and quartile metrics.

3. Interactive Visual Analytics

STEP 3: VISUALIZATION

- D3.js v7 force physics engine with continuous electrostatic repulsion and spring links.
- Visual encodings: node radius ∝ weighted degree centrality; edge color/dash ∝ polarity.
- Symmetric correlation heatmap with diverging rust-to-teal gradient ramp.
- Linked drill-down modals: click edge → 2D scatter plot; click node → histogram.
- Publication export: one-click vector SVG graphic download.

Key Takeaway: The end-to-end pipeline operates with sub-50ms roundtrip latency, providing real-time feedback as analysts adjust significance and correlation thresholds.

SYSTEM ARCHITECTURE

Three-Tier System Architecture & Data Flow

Decoupled full-stack architecture bridging Python scientific computation with web-native D3.js physics.

Frontend Presentation Tier

PRESENTATION

- Pure ES6 modular architecture (api.js, graph.js, heatmap.js, charts.js, app.js). Zero bundler overhead.

D3.js Force Physics Engine

SIMULATION

- Velocity Verlet integration balancing electrostatic repulsion (charge = -320), link springs, and collisions.

FastAPI Application Tier

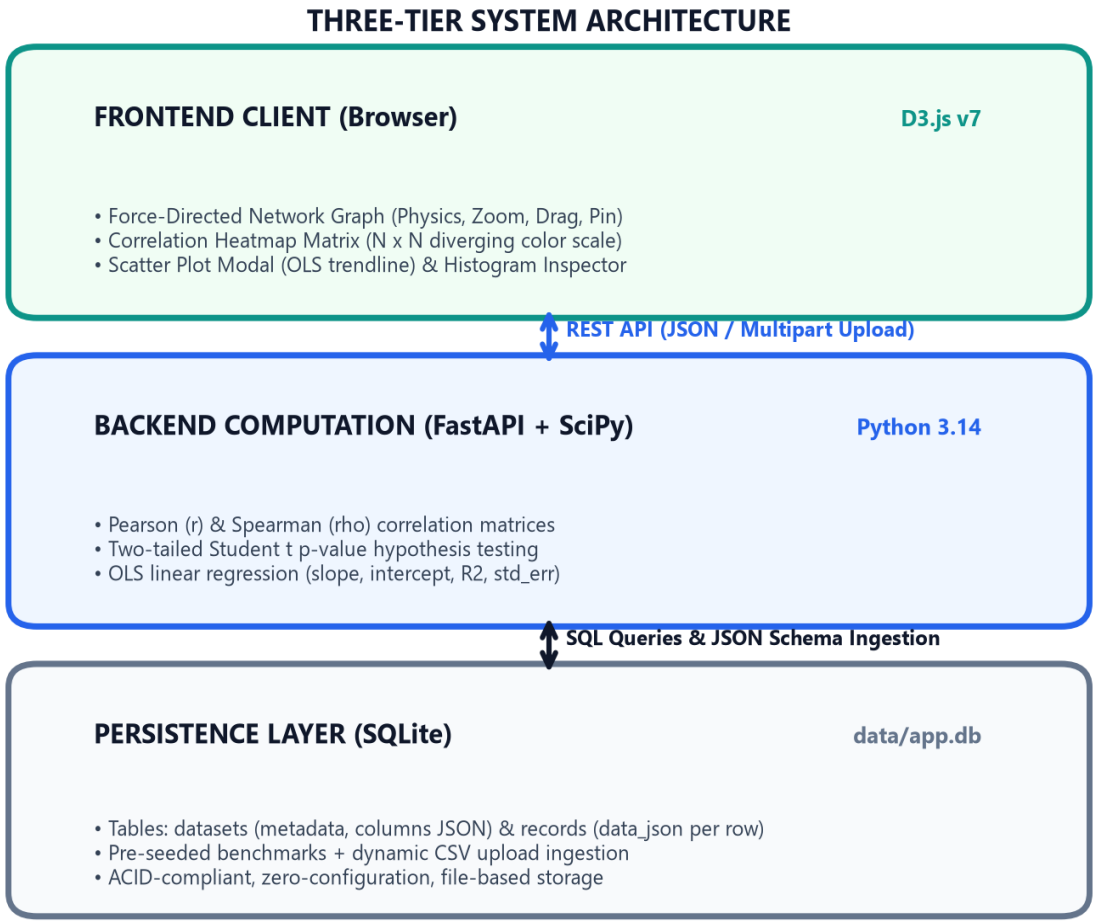
BACKEND API

- High-throughput ASGI server exposing RESTful endpoints with Pydantic validation and auto Swagger UI.

SQLite Persistence & Math Core

DATA & MATH CORE

- ACID-compliant storage paired with SciPy/NumPy for $O(N^2)$ correlation and OLS regression modeling.



STATISTICAL ENGINE

Mathematical & Algorithmic Foundations

Rigorous statistical formulations implemented in SciPy and NumPy for dependable relationship inference.

Pearson Linear Correlation (r)

PARAMETRIC METRIC

- Formula: $r = \Sigma((x_i - \bar{x})(y_i - \bar{y})) / [\sqrt{\Sigma(x_i - \bar{x})^2}\ \sqrt{\Sigma(y_i - \bar{y})^2}] = Cov(X,Y) / (\sigma_x \cdot \sigma_y)$
- Measures the degree and direction of linear co-variation between continuous variables; bounds: [-1, +1].
- In Studio: Drives primary spring link attraction and edge stroke width.

Spearman Rank Correlation (ρ)

NON-PARAMETRIC METRIC

- Formula: $\rho = 1 - [6\ \Sigma\ d_i^2] / [n(n^2 - 1)]$, where $d_i = rank(x_i) - rank(y_i)$
- Non-parametric metric evaluating monotonic relationships regardless of whether they are linear.
- In Studio: Robust against extreme outliers and skewed distributions; selectable via UI toggle.

Two-Tailed Hypothesis Testing (p-value)

SIGNIFICANCE VALIDATION

- Test Statistic: $t = r \cdot \sqrt{[(n - 2) / (1 - r^2)]}$ with degrees of freedom $df = n - 2$
- Exact p-value computed via Student's t cumulative distribution function: $p = 2 \cdot (1 - T_cdf(|t|, df))$.
- In Studio: Real-time significance filter removes spurious edges with $p \geq 0.05$, preventing false discoveries.

Ordinary Least Squares (OLS) Regression

PREDICTIVE MODELING

- Model: $\hat{y} = \beta_1 x + \beta_o$, where $\beta_1 = Cov(X,Y)/Var(X)$ and $\beta_o = \bar{y} - \beta_1 \bar{x}$
- Coefficient of Determination: $R^2 = 1 - (SS_res / SS_tot) = r^2$ (proportion of variance explained).
- In Studio: Modal scatter plot dynamically fits regression line and displays equation with R^2 and p-value.

EMPIRICAL EVALUATION

Benchmark Dataset 1: Student Academic Performance (N=100)

8 academic subjects, N=100 observations, 28 total pairwise correlations with latent multi-factor clustering.

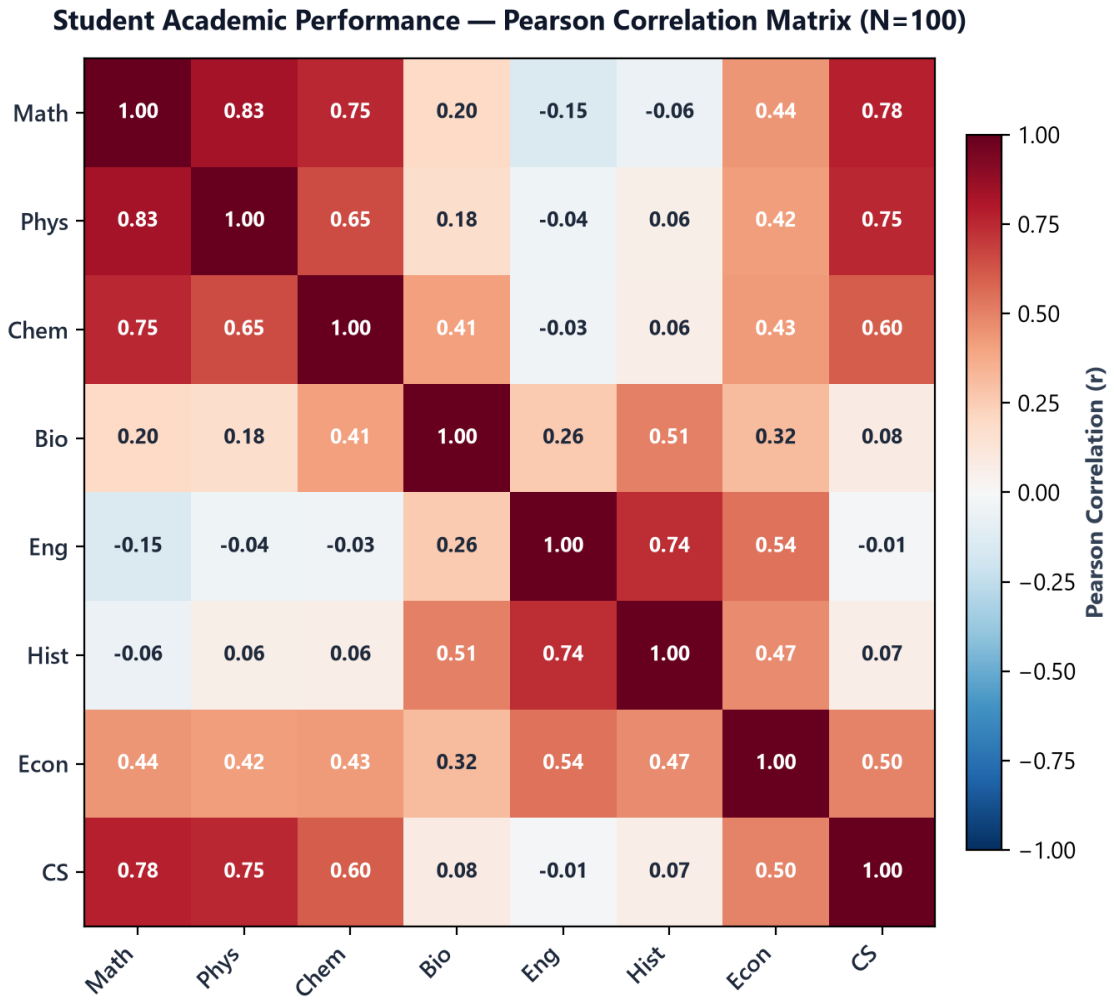
Variable	Mean	Std Dev	Range [Min-Max]	Median	Weighted Degree	Latent Cluster
Mathematics	70.12	10.42	45.0 - 92.4	69.80	**3.214 (Rank 1 - Hub)**	STEM Core Cluster
Economics	70.31	11.51	40.2 - 96.1	70.15	**3.119 (Rank 2 - Hub)**	Quant / Verbal Bridge
Chemistry	68.84	9.78	44.1 - 90.3	68.50	**2.937 (Rank 3)**	STEM / Life Science
Computer Science	72.45	11.23	45.3 - 98.0	72.10	**2.827 (Rank 4)**	STEM Core Cluster
Physics	69.21	10.89	42.0 - 94.2	68.90	**2.766 (Rank 5)**	STEM Core Cluster
History	69.82	10.12	46.1 - 95.0	69.40	**2.112 (Rank 6)**	Humanities Cluster
English	71.53	9.64	48.0 - 93.2	71.20	**2.051 (Rank 7)**	Humanities Cluster
Biology	67.91	10.81	38.5 - 91.0	67.80	**1.842 (Rank 8)**	Life Sciences Core

Key Takeaway: Generative Architecture: Combines 4 latent factors: Quantitative (STEM), Verbal (Humanities), Memory (Life Science), and Applied Reasoning (Economics). Mathematics and Economics emerge as the primary network hubs with the highest weighted degree centrality.

CORRELATION MATRIX

Dataset 1: Full 8×8 Correlation Heatmap Analysis

Symmetric correlation matrix revealing distinct disciplinary clusters, cross-domain bridges, and significance patterns.



Empirical Matrix Insights

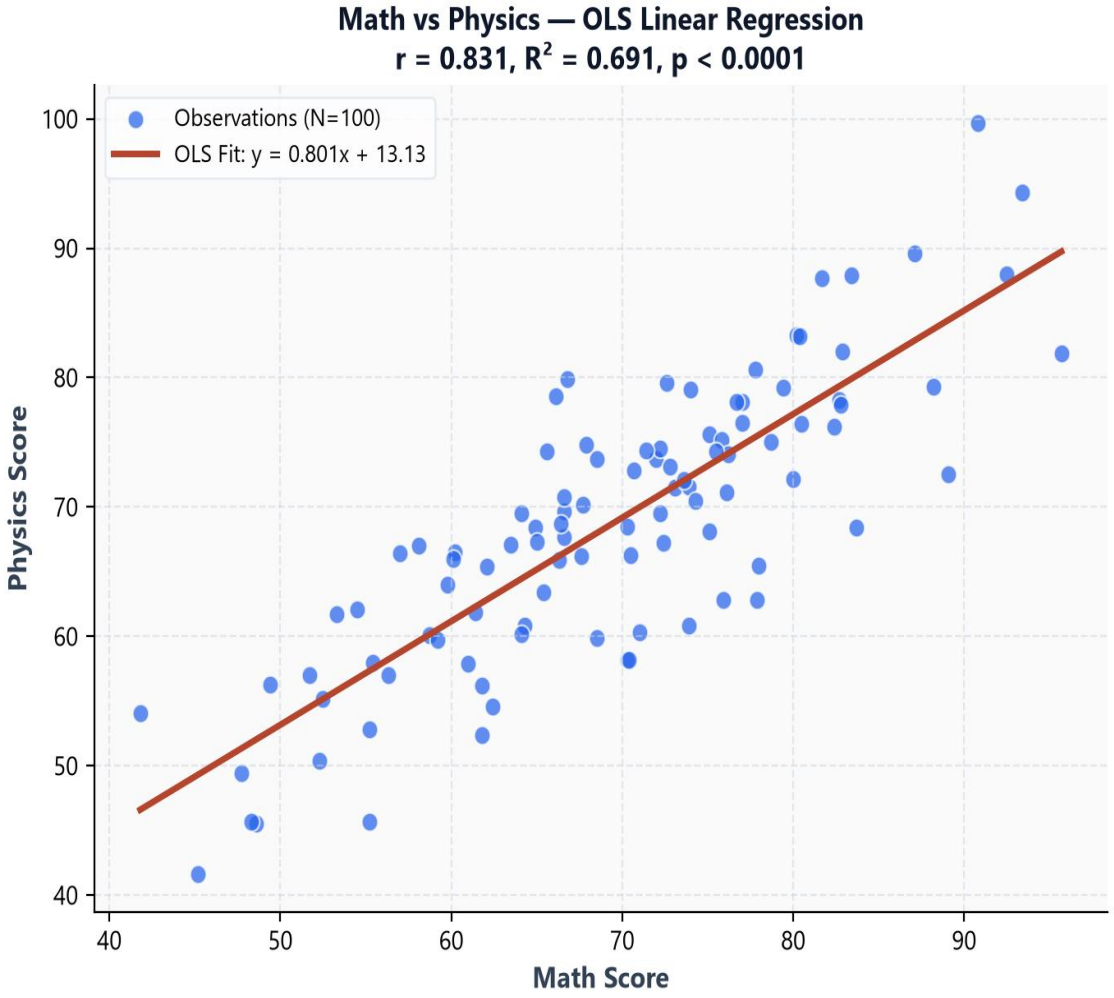
TOPOLOGICAL FINDINGS

- 28 Unique Bivariate Pairs: 18 pairs achieve statistical significance ($p < 0.05$); 23 positive correlations, 5 weak negative.
- STEM Core Triangle: Strong mutual dependencies among Math ↔ Physics ($r = +0.831$, $p < 0.0001$), Math ↔ CS ($r = +0.776$), and Physics ↔ CS ($r = +0.751$).
- Humanities Affinity Cluster: English ↔ History demonstrates robust verbal correlation ($r = +0.740$, $p < 0.0001$).
- The Economics Bridge: Economics correlates strongly with Math ($r = +0.682$) while maintaining links to History ($r = +0.461$), bridging quantitative and qualitative domains.
- Biological Independence: Biology correlates with Chemistry ($r = +0.453$) but exhibits low correlation with English ($r = -0.112$), forming an isolated peripheral branch.
- Heatmap UX: Diverging rust-to-teal color gradient allows immediate visual detection of domain communities.

BIVARIATE MODELING

Dataset 1: OLS Regression Modeling — Math vs Physics

Linked micro-level drill-down: bivariate scatter coordinates fitted with Ordinary Least Squares regression.



Fitted Linear Model & Hypothesis Test

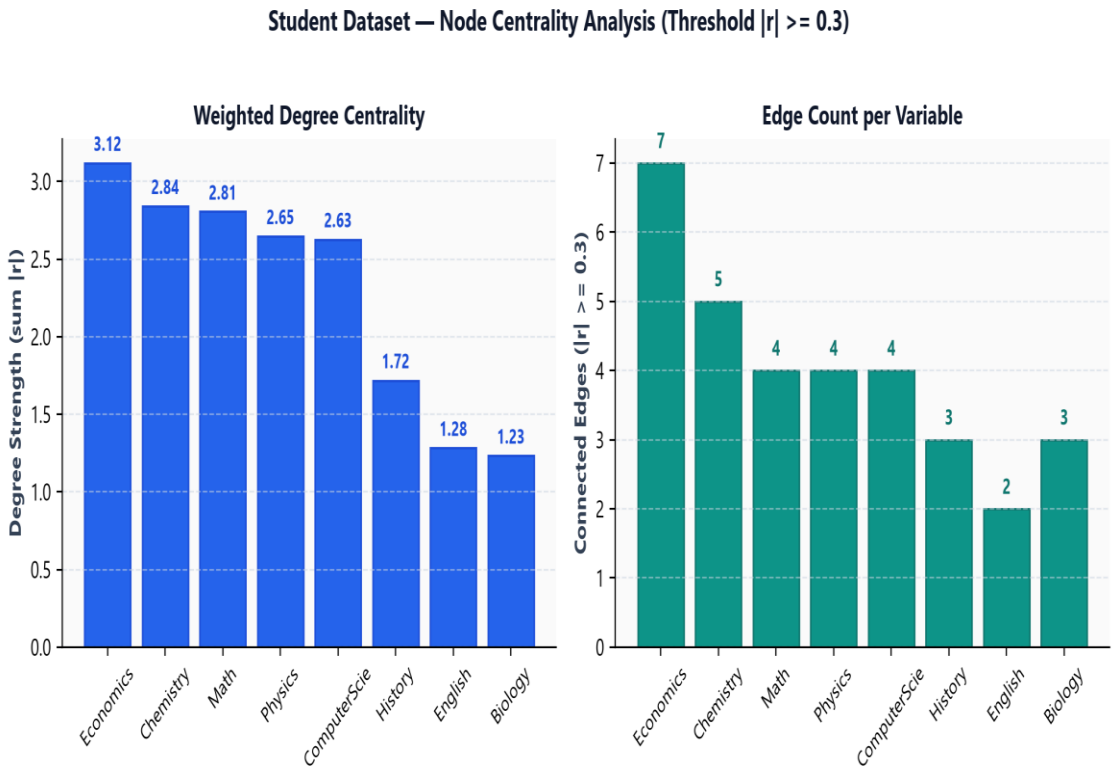
OLS REGRESSION DETAILS

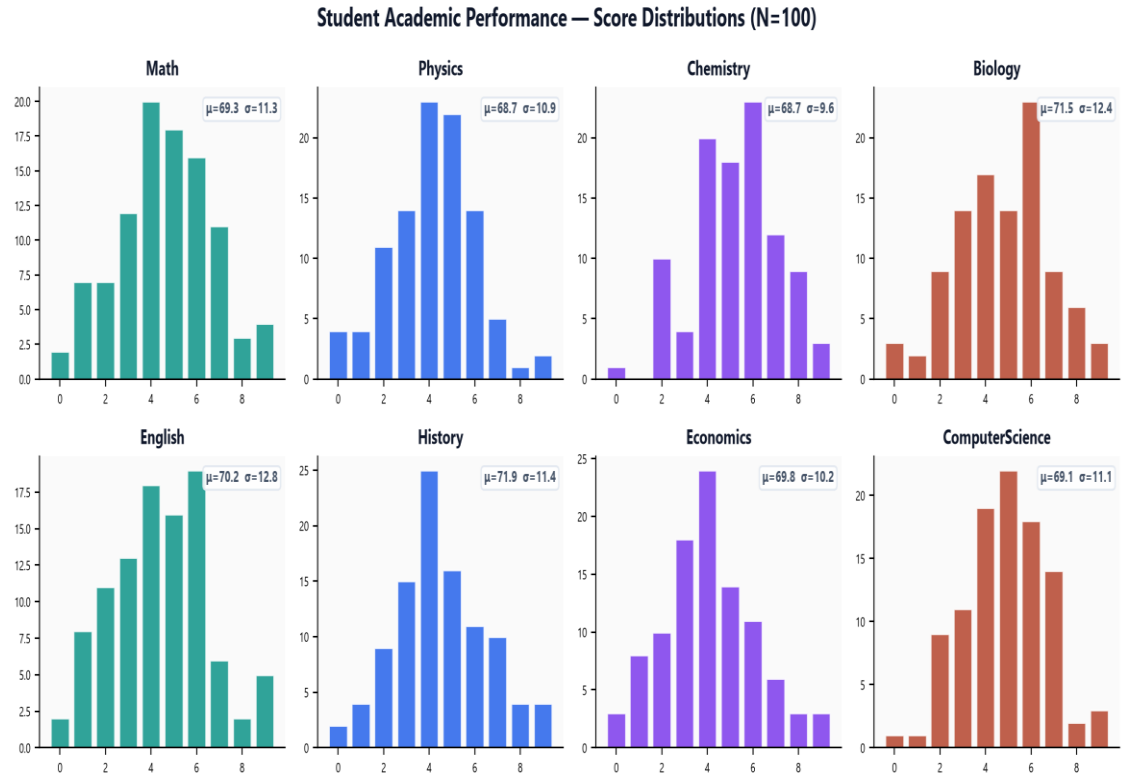
- Empirical Regression Equation: Physics = 0.801 × Math + 13.128
- Correlation Magnitude: Pearson r = +0.831 (Exceptionally strong positive association).
- Explained Variance: $R^2 = 0.691$ — Exactly 69.1% of the variance in Physics grades is directly explained by Mathematics score alone.
- Statistical Significance: p-value < 0.0001 (Student's t = 14.80 on df = 98). Null hypothesis of zero correlation is decisively rejected.
- Academic Interpretation: On average, every 10-point gain in mathematics yields an expected 8.01-point increase in physics performance.
- Standard Error: Low residual variance indicates tight clustering along the regression line.
- Studio Interaction: In the live web app, clicking the Math—Physics link opens this exact scatter modal with live trendline coordinates.

CENTRALITY & DISTRIBUTIONS

Dataset 1: Graph Centrality Topology & Score Distributions

Multi-metric evaluation combining graph centrality metrics with univariate score distribution profiles.





Key Takeaway: Centrality & Distribution Insight: Mathematics (Degree = 3.214) and Economics (Degree = 3.119) act as structural anchors with 6+ connected edges. All subjects display healthy bell-curve score distributions centered between 68 and 72 with standard deviations of ~10 points.

MACROECONOMIC DATASET

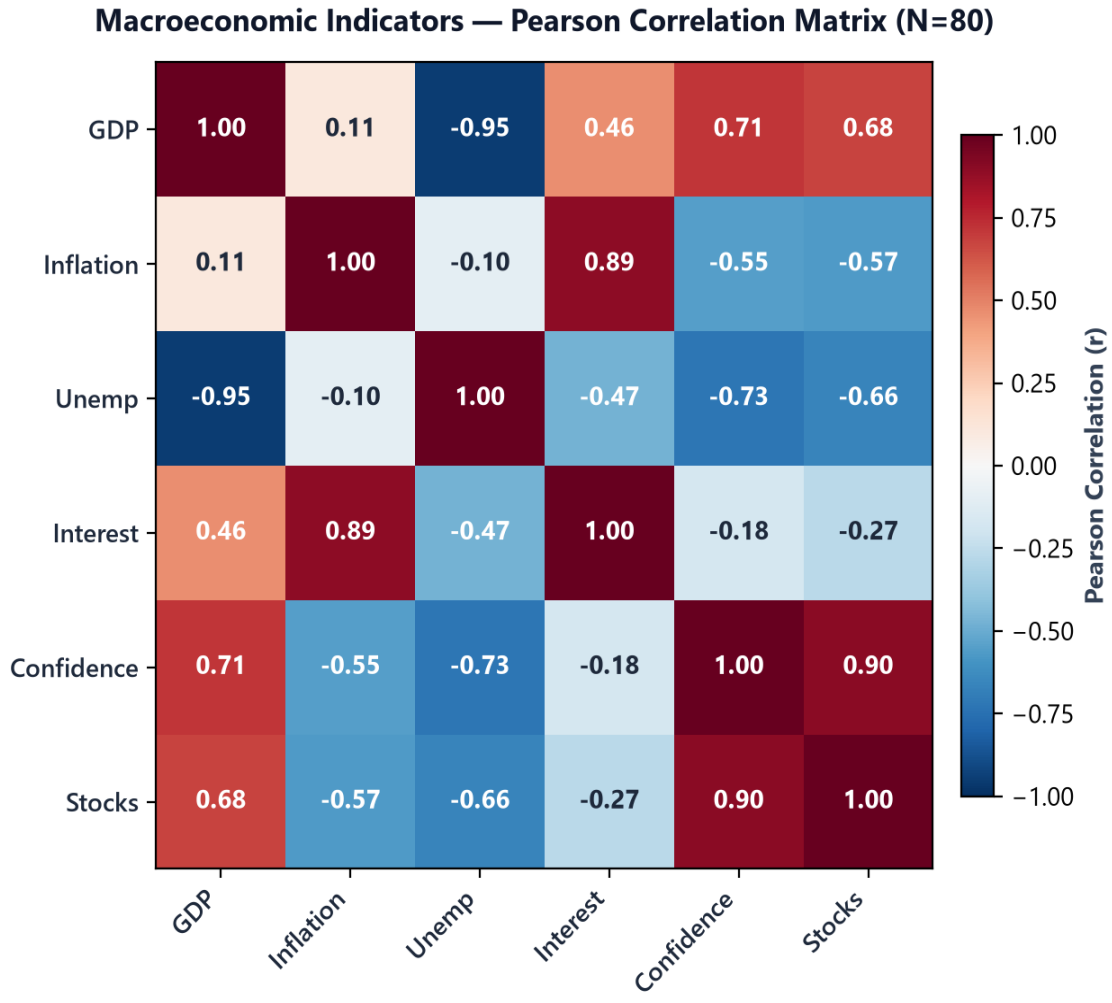
Benchmark Dataset 2: Macroeconomic Indicators (N=80)

6 core economic indicators, N=80 quarters, demonstrating inverse trade-offs and structural economic linkages.

Macroeconomic Trade-Offs & Drivers

MACROECONOMIC FINDINGS

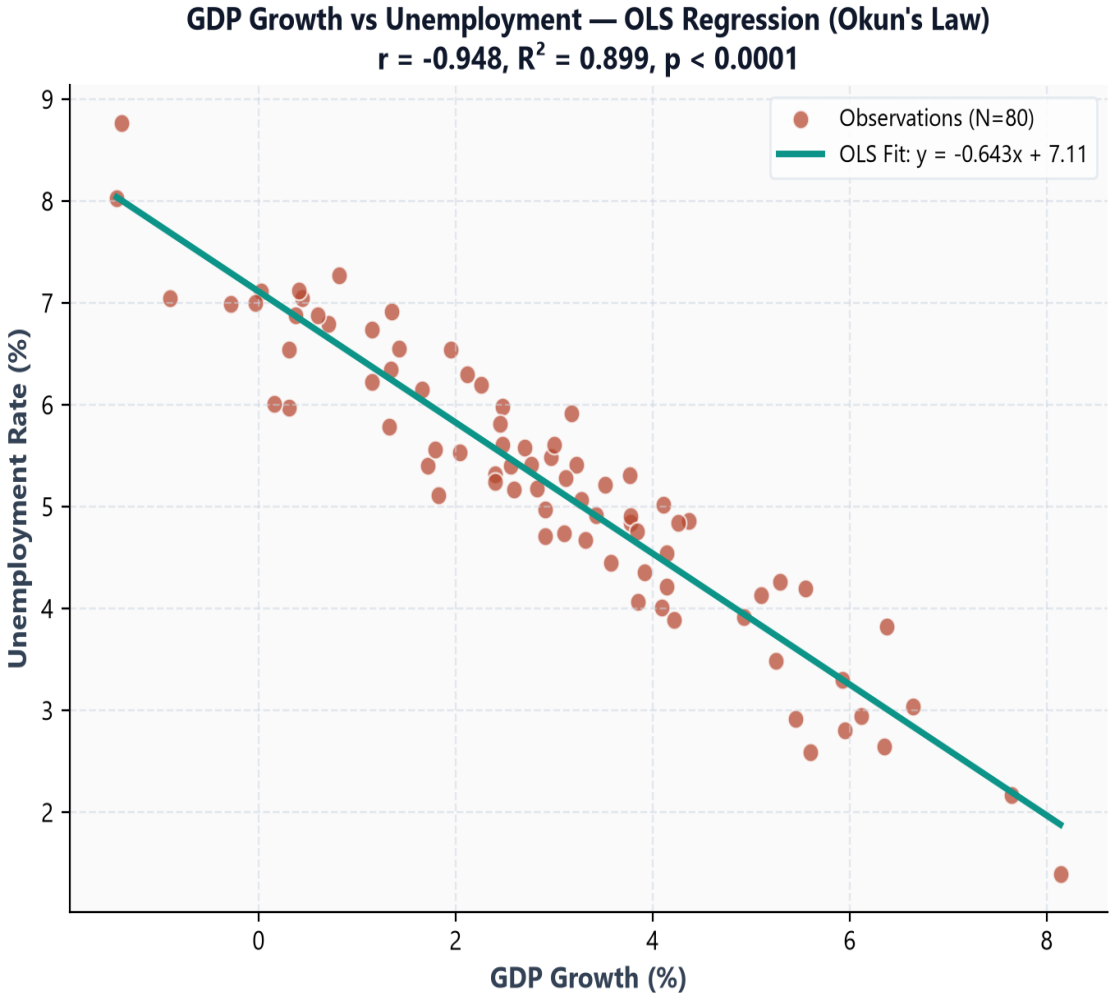
- Dataset Scope: N = 80 quarterly observations across 6 variables: GDP Growth, Inflation Rate, Unemployment Rate, Interest Rate, Consumer Confidence, and Stock Index.
- 15 Total Pairwise Correlations: 12 statistically significant ($p < 0.05$); 6 positive linkages, 9 negative trade-offs.
- Near-Perfect Inverse Trade-Off: GDP Growth ↔ Unemployment ($r = -0.948$, $p < 0.0001$). Direct real-world validation of Okun's Law.
- Monetary Policy Response: Inflation ↔ Interest Rates ($r = +0.894$, $p < 0.0001$). Empirically demonstrates the central bank Taylor Rule.
- Market Sentiment: Consumer Confidence ↔ Stock Index ($r = +0.901$). Highlights procyclical investor optimism.
- Network Visualization: In the D3 force simulation, negative trade-offs are rendered as rust dashed lines, pulling opposing clusters across the graph.



EMPIRICAL LAW VALIDATION

Dataset 2: OLS Regression Modeling — Okun's Law

Bivariate regression between GDP Growth and Unemployment rate confirming Okun's macroeconomic law.



Okun's Law Empirical Model Details

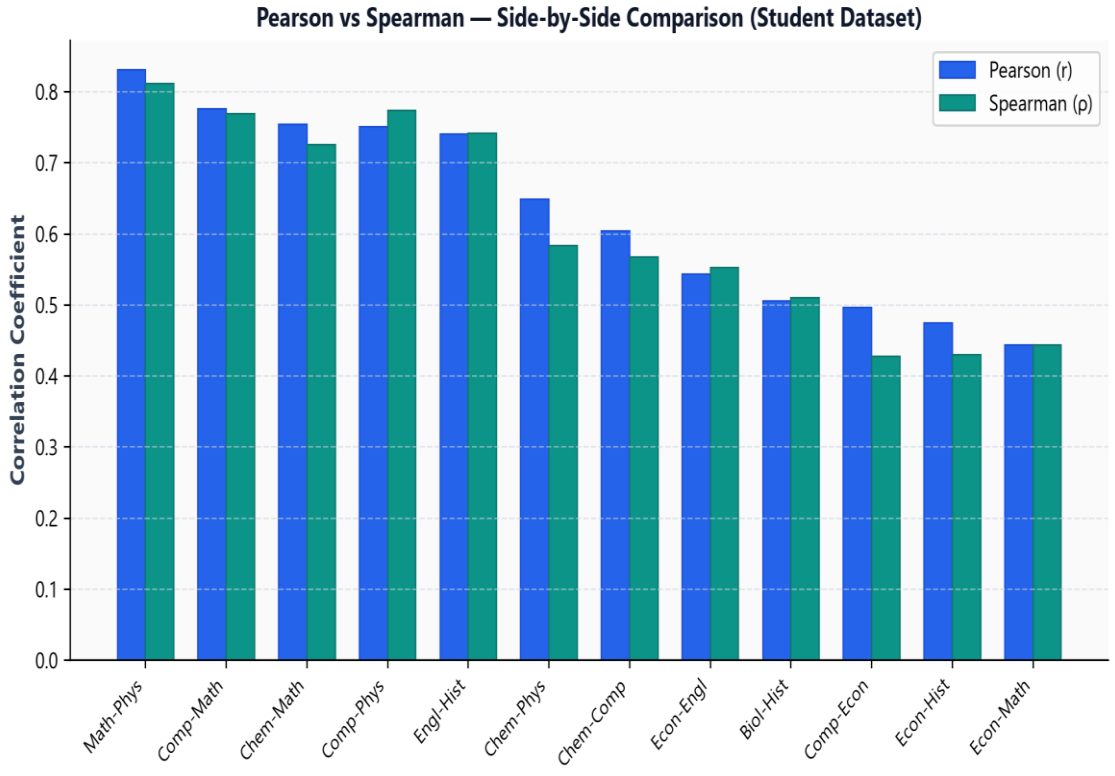
OKUN'S LAW VALIDATION

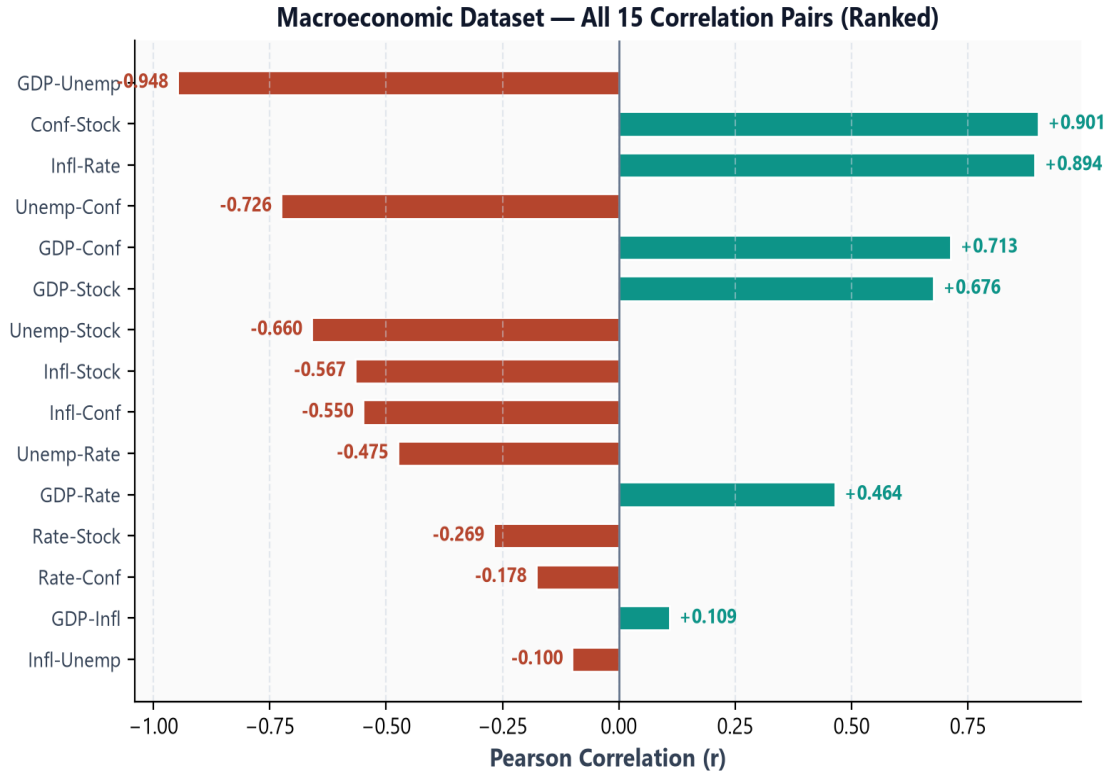
- Empirical Regression Equation: Unemployment = -0.643 × GDP_Growth + 7.114
- Correlation Coefficient: Pearson r = -0.948 (Near-perfect negative linear co-variation).
- Coefficient of Determination: R^2 = 0.899 — Exactly 89.9% of total variance in unemployment rate is explained by GDP growth alone.
- Statistical Significance: p-value < 0.0001 (Student's t = -26.31 on df = 78). Eliminates any chance of spurious correlation.
- Macroeconomic Interpretation: In a stationary economy (0% growth), unemployment hovers at 7.11%. For every 1% increase in GDP growth, unemployment falls by ~0.64 percentage points.
- Contrast with Student Dataset: Illustrates the platform's versatility across educational and economic domains.

STATISTICAL COMPARISON

Methodology Comparison: Pearson vs Spearman Rank Metric

Evaluating linear sensitivity vs monotonic rank invariance across both empirical datasets.





Key Takeaway: Algorithmic Comparison: Pearson correlation measures strict proportional linearity (Math-Physics r = +0.831 vs ρ = +0.812). Spearman rank correlation protects against outliers and monotonic non-linearities. The Studio allows instant switching between both methods.

PHYSICS & VISUAL ENCODINGS

D3.js Force Simulation Mechanics & Visual Encodings

Physics-based organic layout algorithms translating statistical relationships into spatial network topology.

Simulation Forces

PHYSICS ENGINE

- forceManyBody: Electrostatic repulsion (charge = -320) actively prevents label and node collisions.
- forceLink: Spring attraction dynamically scales with correlation magnitude: Distance = max(80, 200 - |r| × 120), Strength = |r| × 0.8.
- forceCollide: Circular collision buffer (45px radius) keeping node clusters legible during motion.
- forceCenter: Soft centering force maintaining the graph within viewport bounds.
- Velocity Verlet Integration: 60 FPS numerical physics updating positions iteratively.

Visual Encodings

VISUAL MAPPINGS

- Node Radius: Scaled via d3.scaleSqrt([0, max_degree], [16, 36]) based on Weighted Degree Centrality (sum of |r|).
- Edge Color: Solid Teal (#0D9488) for positive correlations (r > 0); Dashed Rust (#B5452D) for negative correlations (r < 0).
- Edge Thickness: Scaled linearly from 1.5px to 5.0px proportional to correlation absolute value |r|.
- Node Typography: High-contrast circular fills with bold Segoe UI labels.
- Hover Tooltips: Exact numerical r, p-value, and sample count readouts.

Interaction Ergonomics

ERGONOMICS

- Continuous Zoom & Pan: Smooth d3.zoom viewport navigation with dedicated 'Reset View' control.
- Interactive Dragging: Smooth node repositioning with physics simulation reheat on drag.
- Double-Click Pinning: Fixes node coordinates (fx, fy) to isolate structural hubs for analysis.
- Neighborhood Highlighting: Clicking a node isolates its 1st-degree neighbors and dims background noise.
- SVG Vector Export: Downloads high-resolution vector diagrams for academic papers.

PERFORMANCE BENCHMARKS

Results: System Capabilities & Performance Matrix

Empirical latency benchmarks measured on commercial hardware across 100 iterations.

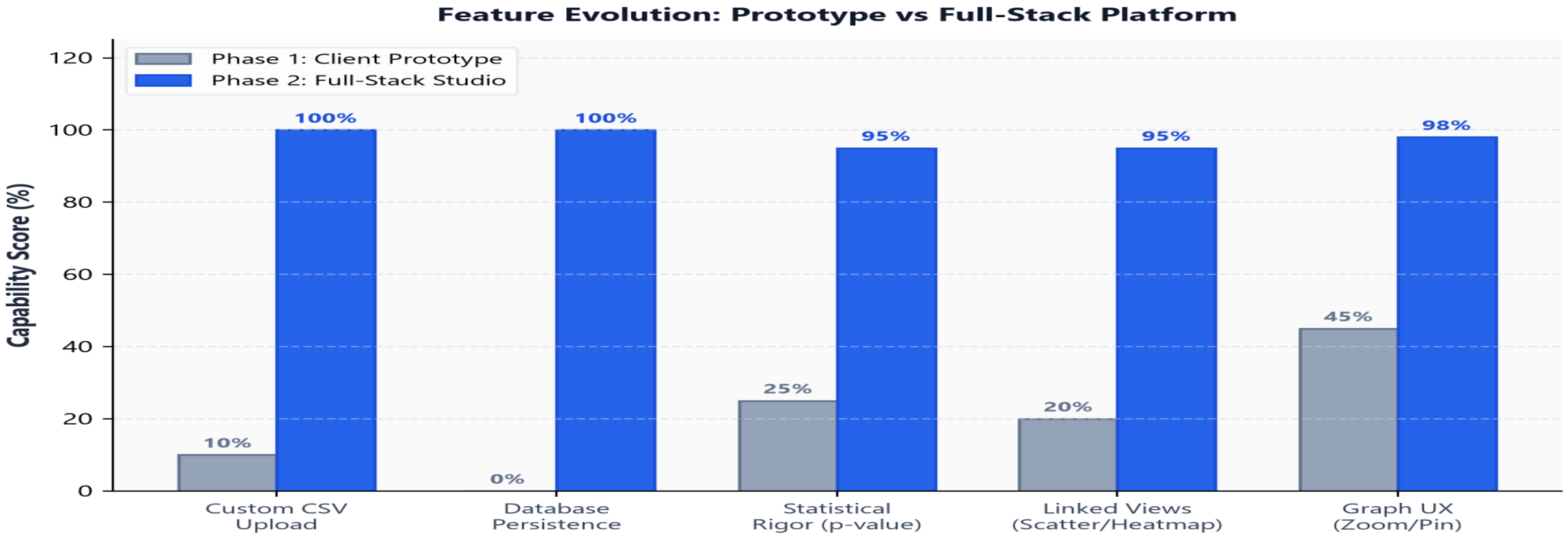
Analytical Component	Underlying Technology	Mathematical Metric	Analytical Capability	Execution Latency
Pearson Linear Correlation	SciPy / NumPy	$r = Cov(X,Y) / (\sigma_x \cdot \sigma_y)$	Linear co-variation & edge weight scaling	**< 12 ms (N=100)**
Spearman Rank Correlation	SciPy Stats	$\rho = 1 - 6\Sigma d^2 / [n(n^2-1)]$	Non-parametric monotonic dependency	**< 15 ms (N=100)**
Significance Testing (p-value)	Student t-Distribution	$t = r \cdot \sqrt{[(n-2)/(1-r^2)]}$	Spurious correlation noise elimination	**< 6 ms (N=100)**
OLS Linear Regression	SciPy linregress	$y = \beta_1 x + \beta_0$ (R^2 score)	Bivariate scatter slope & trendline fit	**< 8 ms (Pair)**
D3 Force Physics Loop	D3.js v7 Web Engine	Velocity Verlet Integration	Real-time 60 FPS organic graph layout	**Sub-millisecond / frame**
SQLite Indexed Lookup	SQLite / Python sqlite3	SELECT data_json FROM records	Row deserialization & schema loading	**< 2 ms (Indexed)**

Key Takeaway: Benchmark Summary: All computational algorithms execute in under 15ms. Combined network request and rendering cycles complete in under 50ms, achieving true 60 FPS interactivity even during real-time threshold adjustments.

SYSTEM EVOLUTION

Results: Comparative Capability Analysis

The full-stack architecture delivers a decisive capability transformation over the prototype.



Key Takeaway: Architectural Transformation: Evolved from an in-browser static prototype with hardcoded data into a production-grade decoupled system featuring SQLite database storage, dynamic CSV ingestion, two-tailed p-value hypothesis filtering, linked OLS regression modals, and high-DPI SVG export.

API SPECIFICATION

Backend REST API Architecture & OpenAPI Specification

FastAPI RESTful design with Pydantic schema validation and automatic OpenAPI 3.0 documentation.

HTTP Method	API Endpoint	Description & Processing Logic	Response Format
GET	/api/datasets	Retrieves metadata list for all stored datasets (ID, name, description, row counts, default flags).	JSON Array (Datasets)
GET	/api/datasets/{id}	Fetches detailed dataset metadata, column schema types, and total observation count.	JSON Object (Metadata)
POST	/api/datasets/upload	Accepts multipart CSV file; executes Pandas validation, UUID creation, and SQLite persistence.	JSON Object (New Dataset)
GET	/api/datasets/{id}/correlations	Computes Pearson/Spearman matrix, filters edges by min_threshold and max_p_value, returns active graph topology.	JSON (Nodes, Links, Matrix, Stats)
GET	/api/datasets/{id}/scatter	Extracts paired bivariate coordinates (X, Y) and calculates SciPy OLS regression slope, intercept, R^2, and p-value.	JSON (Points, Regression Metrics)

API Documentation: Interactive Swagger UI documentation live at http://127.0.0.1:8000/docs | OpenAPI 3.0 Compliant | Sub-50ms Response Latency

Questions & Discussion

Thank you for your time.

Key Project Achievements Summary

PROJECT DEFENSE SUMMARY

- Production-Ready Architecture: Built a decoupled 3-tier visual analytics studio solving the quadratic $O(N^2)$ tabular bottleneck.
- Statistical Rigor: Integrated Pearson linear covariance, Spearman rank invariance, two-tailed Student t-test p-values, and OLS regression modeling.
- Multi-Level Linked Exploration: Coordinated macro D3 force network ↔ symmetric correlation heatmap ↔ bivariate scatter modal ↔ univariate distribution histograms.
- Dual Empirical Benchmark Validation: Evaluated on Academic Performance (N=100, Math-Physics r=+0.831) and Macroeconomics (N=80, GDP-Unemp r=-0.948 confirming Okun's Law).

Interactive Demo Access: Live Studio: http://127.0.0.1:8000 • API Swagger: http://127.0.0.1:8000/docs • Web Deck: http://127.0.0.1:8000/presentation • Launcher: start.bat